

Automobile Data – Preprocessing & Visualization Project

Objective

Leverage Python and visualization tools to clean, transform, and analyze automobile data to uncover actionable insights about vehicle pricing, brand trends, and performance metrics.

Problem Statement

The automobile industry generates large volumes of raw data related to vehicle specifications, brand performance, and market pricing. However, this data often contains missing values, inconsistencies, and skewed distributions that hinder direct analysis. The goal is to:

- Clean and preprocess the dataset for analysis.
- Explore pricing patterns and feature relationships.
- Understand how different attributes (e.g., brand, engine size, fuel type) impact car pricing and efficiency.
- Build meaningful visualizations to support data-driven insights.

Dataset Overview

[Click here for the dataset](#)

The dataset contains automobile specifications including:

- **Vehicle Specs:** Make, Body Style, Engine Size, Horsepower, Fuel Type
- **Efficiency Metrics:** City & Highway MPG
- **Price Data:** Price, Curb Weight, Length, Width
- **Drive Info:** Drive Wheel, Number of Doors

Key Analysis Questions

Vehicle Pricing Trends

1. What are the major factors influencing vehicle price?
2. How does brand reputation (e.g., BMW vs. Toyota) correlate with pricing?

Engine & Performance Insights

1. Do larger engines and higher horsepower lead to higher prices?
2. How do different fuel types affect performance and cost?

Design & Drive Style Analysis

1. Are certain body styles or drive types more expensive?
2. What impact do dimensions (length, width, weight) have on pricing?

Suggested Visualizations

1. **Correlation Heatmap** – Price vs. Numerical Features
2. **Bar Charts** – Average Price by Brand or Fuel Type
3. **Boxplots** – Price Distribution Across Body Styles
4. **Scatter Plots** – Horsepower vs. Price, Engine Size vs. MPG
5. **Pairplot** – Multivariate Pattern Exploration